VAMSI DEVARAPALLI

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Technical Skills

- **Programming Languages:** Python, SQL, Linux Shell Scripting
- **Databases & Data Warehouses:** Apache Hive, PostgreSQL
- **Message Queue:** Confluent Kafka
- **Distributed Computation Frameworks:** Apache Hadoop, Databricks (Apache Spark)
- **Workflow Management:** Airflow
- **AWS Services:** S3, Lambda, Redshift, Glue, DynamoDB, IAM, Kinesis, RDS, Athena, EMR
- **Azure Services:** Azure Synapse analytics, Azure Blob Storage, Azure Data factory, Azure Datalake storage (ADLS)

Experience

Vectra Automation Inc., — **June 2023 – Present**

Data Engineer — *Chennai, India*

➢ **SCD (Type 2) Data ingestion pipeline using Databricks**

Tech Stack*: PySpark, Python, Databricks, AWS S3, AWS IAM*

- Authored PySpark script to transform and ingest the latest (fact) data into the delta table.
- Configured the Databricks workflow to instantiate pipeline runs in a 100% automated manner on a daily basis as per the scheduled time.
- Leveraged Python (pydeequ package) to author data quality checks, ensuring 100% quality data ingestion.
- Setup an e-mail alerting system to monitor the pipeline 24/7, providing instant status updates on pipeline runs.

➢ **Automated pipeline for incremental data ingestion using AWS Glue**

Tech Stack: Spark, S3, Glue, Redshift, Event Bridge, SNS, IAM

- Leveraging Glue's Job Bookmarking feature, configuring the crawler to accurately identify the schema of newly arrived data to ensure 0% redundant computation.
- Ensured only quality data is ingested with 100% consistency using glue's data quality checks and the results are published into an SNS topic notifying the subscribers through e-mail.
- Succeeded records are routed to redshift table and failed records are dumped into S3 bucket, ensuring 0% data loss

➢ **Airflow orchestrated pipeline for data warehouse management**

Tech stack*: GCP Dataproc, Composer, Cloud Run Functions, Cloud Storage buckets (GCS), Python, Apache Hive*

- Configured a Cloud Run Function with an average execution time of 800 ms per request to trigger the Airflow DAG (in composer environment) upon receiving the daily data from an upstream source.
- Ingested the input data into a partitioned Hive table hosted on a Dataproc cluster.
- Engineered the pipeline to prevent unnecessary triggers upon receiving bad data.

Personal projects

➢ **Near real-time streaming data pipeline using AWS Kinesis**

Tech Stack: *Python, AWS S3, AWS EventBridge, AWS Kinesis, AWS DynamoDB, lambda*

- Connected seamlessly between DynamoDB and Kinesis through Event Bridge for event-driven integration, facilitating the capture and processing of stream data.
- Implemented 100% effective Change Data Capture (CDC) in DynamoDB, enabling streamlined tracking.

Education

Vasireddy Venkatadri Institute of Technology (VVIT) — **Apr 2019 – Apr 2023**

Bachelor of Technology in Civil Engineering — *Guntur, Andhra Pradesh*